

C-Doppelstander – German occupation ensign

A technical documentation of the Allied control ensign used by German merchant shipping, 1945–1951



C-Doppelstander swallowtail ensign (German merchant shipping 1947–1951), FG-047 on the right, FG-048 on the left

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Executive Summary

The Allied C-Doppelstander: Signal-Flag Geometry and Operational Practice

This report examines the design, construction, and operational use of the Allied **C-Doppelstander**, a swallow-tailed control flag used on German merchant vessels during the Allied occupation following World War II. Analysis of surviving physical examples, photographic documentation, and Allied Control Council regulations reveals several important observations that contradict existing literature.

First, the underlying geometry of the flag corresponds to the **International Code of Signals flag “C”**, which follows an approximate **5:6 proportion** typical for maritime signal flags. Surviving examples consistently reflect this geometry rather than the **2:3 or 3:5 proportions** normally associated with naval ensigns, suggesting that the occupation flag was derived directly from a standard signal flag rather than newly designed.

Second, the swallow-tail itself appears to have been produced by **cutting a signal flag to create a double-pennant form**, preserving the original signal-flag proportions. This explains why surviving examples maintain signal-flag geometry despite the later schematic depiction in Allied Control Council documentation.

Third, a discrepancy exists between the regulatory illustration and physical evidence. The Allied Control Council regulation illustrates a **90-degree tail cut**, while surviving examples consistently display a **much wider tail angle of approximately 110 degrees**, corresponding to typical swallow-tail construction used in maritime signal flags.

Fourth, wartime evidence indicates that the signal flag **“C”** had already been referenced in Allied communications prior to the occupation. A broadcast in **1943** referring to Allied control signals directed toward Italian shipping suggests that the concept of using signal-flag identifiers for control or compliance may have been considered before the formal occupation regulations were issued.

Finally, the presence of swallow-tail variants among several signal flags raises the possibility that the C-Doppelstander formed part of a broader signalling logic derived from the **International Code of Signals**. In this interpretation, the flag represents a practical adaptation of an existing signal flag, modified to create a distinctive control signal rather than a newly designed occupation emblem, similar in principle to swallow-tail adaptations seen in other signal-flag contexts.

Taken together, the available evidence indicates that the Allied C-Doppelstander should be understood as an **operational modification of the signal flag “C” rather than a newly designed occupation flag**, reflecting established maritime signalling practices rather than the simplified schematic form shown in post-war regulations.

1) Introduction / Research questions

Flag ID: FG-047 & FG-048

After the German capitulation in May 1945, all national insignia were banned, and German merchant vessels were ordered to operate under Allied control. One of the most obscure results of this policy was the *C-Doppelstander* — a modified version of the International Signal Flag “C” used as a temporary ensign for German shipping until 1951. Despite its importance in the early post-war period, very few examples survive, and its exact proportions, construction details, and historical rationale remain poorly understood.

This study brings together six independent references — including two maker-stamped cloth examples, archival photographs, a museum piece, and period film footage — to reconstruct how the flag was made, how it was used, and why the Allies chose this particular design.

The study asks four practical questions about the *C-Doppelstander* (the Allied occupation ensign used by German merchant shipping after 1945):

- **(1) Authenticity.** Given how few survive, are our two pieces authentic period work?
- **(2) Ratio and size.** What were its *real* proportions and size class in use (ratio and nominal sizes)?
- **(3) Origin and manufacture.** Was it in practice derived from a standard *signal flag C* blank?
- **(4) Why “C” — and the link to D / E.** Why was the letter C chosen (and not A or B) — and how does this relate to the later *D* and *E* occupation flags?

We build the case from **six independent reference points**:

- **FG-047** — our maker-stamped cloth example.
- **FG-048** — our second, matching maker-stamped example.
- **DDG Greiffenfels period photograph** with the ensign in service.
- **Biegler’s published photograph** (Gr. 1 C-Doppelstander) confirming the signal-flag basis.
- **Bonn museum reference** (collection documentation/holding relevant to the type).
- **Contemporary film footage** showing the ensign under way.

Across these references we test two linked hypotheses: first, that workshops and crews reused *signal C* stock ($\approx 5:6$ family) rather than manufacturing a new rectangular standard (e.g., 2:3 or 3:5); second, that the swallowtail was cut to a broad, practical wedge ($\approx 106\text{--}110^\circ$) rather what is shown in Annex A to Allied Control Council Law No. 39.

Method in brief

- **Object work:** fabric, stitching, hardware, and hoist markings on FG-047/048.
- **Origin question:** why “C” in 1945, and how the logic continues to “D” (Ryukyu) and “E” (Japan).
- **Regulatory baseline:** 9 May 1945 order and ACC Law No. 39 (schematic only, no cm tables).
- **Measurements:** preserved sizes, ratio checks, and a photogrammetric cross-check against *Greiffenfels*.
- **Reference material:** Biegler’s published image; museum documentation; film stills.
- **Authentication track:** maker stamps, materials/threads, and period fittings.

Why this matters

Online summaries often assume neat 2:3 or 3:5 rectangles. Our cloth, photos, and film suggest a different, signal-driven reality. By anchoring the study in six independent references and the tests we run on them, we show the physical standard actually used at sea and the operational reasoning behind choosing “C” (and later “D”/“E”).

Preview of findings

All six anchors converge on a signal-flag body ($\approx 5:6$) with a broad, traditional swallowtail cut; our two maker-stamped flags present as period, working originals. The detailed evidence and caveats follow in Sections 3–9.

2) Overall facts of our specimens

- **Designation:** C-Doppelstander (Erkennungsflagge deutscher Handelsschiffe)
- **Period in force:** Order of 9 May 1945 (Allied war diary – provisional practice); formally in force 17 January 1947 – 23 February 1951.
- **Manufacturer:** Düsseldorfer Fahnenfabrik — Düsseldorf workshop, stamped on hoist.
- **Markings:**
 - Maker's stamp: "Düsseldorfer Fahnenfabrik".
 - Size code: "Gr. 2" (Größe 2).
 - Flag code: "C" (International Signal Flag for "Affirmative")
- **Flag code:** "C" — the international signal flag C (Signal Flag C), cut as a swallowtail to form the C-Doppelstander ("double-stander"; German occupation ensign).
- **Also referred to as:** C-Doppelstander; C-Pennant; double-stander; occupation ensign; Allied-occupation German flag; Signalflagge C (swallow-tailed); Erkennungsflagge.
- **Material:** Wool bunting; heavy canvas hoist.
- **Hardware:** Period-made Inglefield-style clips (German naval terminology – *Brummelhaken*).
- **Measured size:**
 - FG-047: 93 × 112 cm (measured along the upper edge: hoist → upper fly corner).
 - FG-048: 96 × 115 cm (measured along the upper edge: hoist → upper fly corner).
 - *Minor variation due to shrinkage, wear, and hand-cut swallowtail.*
- **Form:** Swallowtail ensign; based on a rectangular signal-flag field before the cut.
- **Ratio:** 5:6 signal-flag family; see Section 6.
 Maker's stamp on both hoists identifies Düsseldorfer Fahnenfabrik (DFF). The workshop's continuity into the early 1950s is set out in Section 8, providing the archival basis for this attribution.

3) The C-Doppelstander — legal and practical use

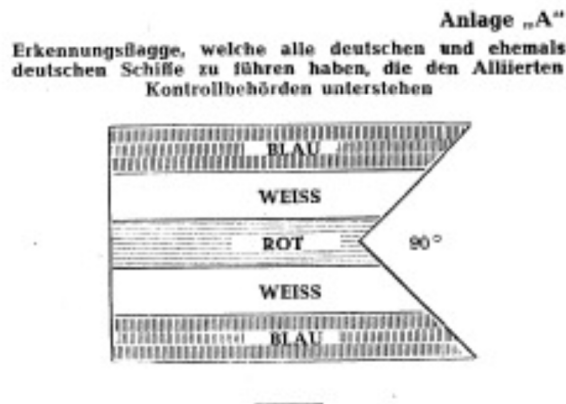


Modern schematic of the C-Doppelstander ensign.

Source: [Wikipedia – Erkennungsflagge für deutsche Handelsschiffe](#),
CC BY-SA 3.0.

After the German capitulation of 7–8 May 1945, Reich flags were banned as part of denazification, and German shipping operated only under Allied control. An Allied war diary entry of 9 May 1945 directed German war and merchant vessels “to fly at the peak International Code Flag C with a triangle cut from the fly thus transforming it into a burgee.” [War Diary, 9 May 1945 \(p. 215\)](#).

This provided a provisional identification ensign under Allied supervision, not a formally standardised flag.



Original Annex A schematic from Allied Control Council Law No. 39: signal flag C with a 90° cut (legal baseline).

Source: [Deutsche Nationalbibliothek – Law No. 39 Annex A](#).

It was only with Allied Control Council Law No. 39 (issued 12 Nov 1946, effective 17 Jan 1947) that the C-Doppelstander was codified in regulation as the official German merchant ensign. The law’s Annex A shows only a schematic sketch—“C” plus a 90° cut—without dimensions or ratios (see Section 6). Any numerical reconstructions of proportions must therefore be regarded as approximate.

In practice, however, the flag’s operational life was limited. With the Basic Law of 23 May 1949, black-red-gold was constitutionally recognised as the flag of the Federal Republic. From that point, German shipping had no incentive to continue with the C-Doppelstander. The ensign remained in formal regulations until 1951, but its effective service life was essentially 1945 – 1949.

Open question: while the purpose and duration of the C-Doppelstander are well documented, its *selection* remains puzzling. Why the letter “C” — and not another signal flag?

No Allied record explains this, yet the same pattern later reappeared in the D- and E-occupation flags of the Ryukyu Islands and Japan. This recurring logic suggests the choice was neither random nor symbolic, but rooted in Allied signal practice.

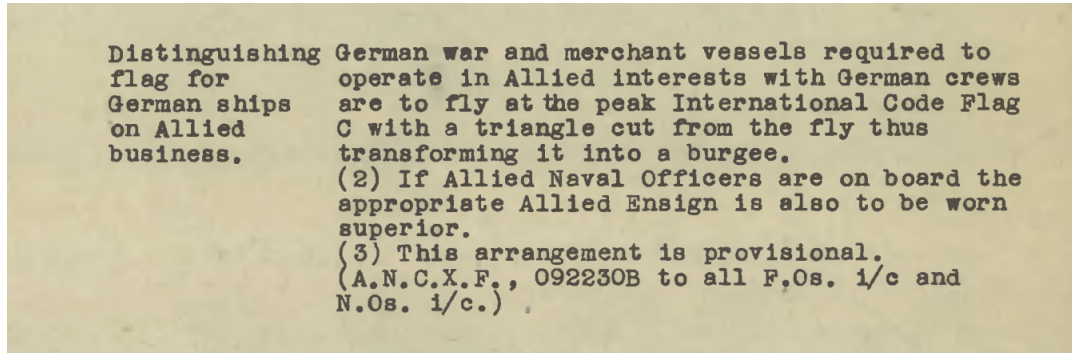
Section 3a therefore explores the earliest known uses of signal flag C and the possible rationale behind its adoption as an occupation ensign.

3a) Why “C”? — early rationale and possible origins

This synthesis outlines the working hypothesis for why the International Code flag “C” (Charlie) was selected for use on German vessels after May 1945 and how it later evolved into the broader Allied occupation-flag system. The evidence below remains under review pending final archival confirmation.

The Allied order of **9 May 1945** instructed all German and ex-German ships “to fly at the peak International Code Flag C with a triangle cut from the fly.”

No reason for choosing the letter C was recorded, yet several operational and historical factors suggest the decision followed established signalling logic rather than improvisation.



Extract from *Allied Naval War Diary*, 9 May 1945:

“...to fly at the peak International Code Flag C with a triangle cut from the fly.”

Source: [Extract from the Royal Navy War Diary – May 1945, p. 215](#) (UK National Archives / Naval Historical Branch).

Practical availability and adaptation.

Before the war, all internationally trading German merchant ships were required to carry full sets of International Code of Signals flags from suppliers such as DFF and Schreiber. During the war, most of the fleet was militarised under the *Reichskommissariat für Seeschifffahrt* and operated under the *Flaggenbuch* 1939 system used by the Kriegsmarine.

By May 1945, only a few civilian vessels still carried ICS sets, yet this was enough to make the Allied order immediately practicable — crews could simply cut the fly of an existing C flag.

In surrendered ports such as Kiel, Flensburg, and Bremerhaven, Allied Control Officers likely authorised or issued replacement ensigns and signal flags for cleared German vessels — a process described in surviving **British Naval Control Service** correspondence (British Library, Add. MS 71786 ff., 1946–47).

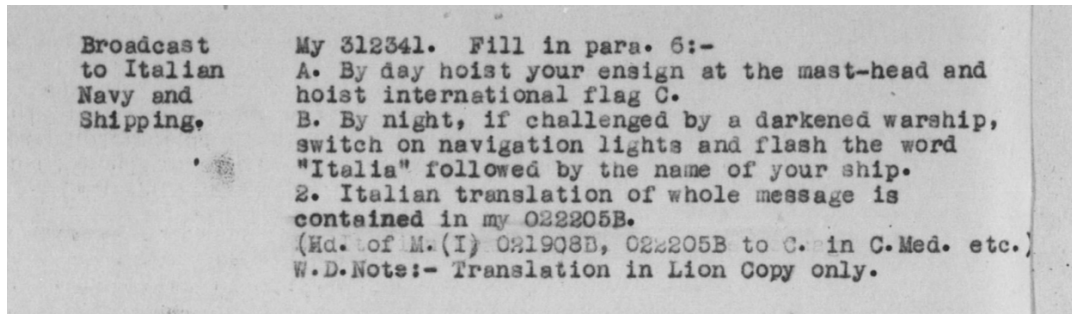
Licensed German workshops soon resumed production under supervision, as indicated by surviving Düsseldorf *Fahnenfabrik* stamps. The preserved DFF-marked C-Doppelstander examples therefore represent the formalised post-surrender production phase — directly evolved from the 1945 field practice.

Origins and symbolic rationale — why “C”?

- 1. **Pre-existing wartime use.**

In the **August 1943 Italian armistice test broadcast**, Allied instructions ordered surrendered Italian ships to hoist both their national ensign and the *Charlie* flag to denote Allied supervision. Although apparently a trial procedure, the instruction followed the standard meaning of “C” in the *International Code of Signals* — an acknowledgment or affirmative response — effectively *I comply*.

This precedent likely influenced the 1945 selection.



Allied naval test broadcast, August 1943:

“...Italian ships under Allied control to hoist International Code Flag C and their national ensign.”

Source: [NARA RG 38 – Combined Chiefs of Staff Signal Summary Vol. IV \(1943\)](#).

- **2. Continuation and necessity.**

When applied to Germany in 1945, the same signal concept was retained but modified with a **swallowtail cut** to prevent confusion with the square signal flag. With no national ensign left to hoist, the C flag itself became the sole identifier — neutral, distinct, and lawful under Allied supervision.

- **3. Sequential logic.**

Within the *International Code of Signals*, flags **A** and **B** already carried swallowtail cuts, leaving **C** as the next logical symbol. The result met legal ensign requirements while avoiding banned insignia — a pattern that soon reappeared elsewhere. As rumours suggest, flags **A** and **B** had already seen limited experimental use in the Pacific (around the Bikini Atolls), which makes it plausible that the idea simply evolved: combining the familiar swallowtail form of **A/B** with the existing **C** flag already used in Allied signal procedures for identifying controlled or surrendered ships.

The **C-Doppelstander** would thus represent a natural fusion of these two precedents.

No primary documentation yet confirms A/B usage; the connection remains circumstantial.

- **4. Neutral and non-salutable character.**

As stated in *Allied Control Council Law No. 39*, the occupation ensign was to be regarded as a *neutral flag* - it was not to be saluted, honoured, or treated as a national colour. This denazification clause reflected the Allied aim to remove all symbolic loyalty from state emblems while maintaining seagoing identification.

Thus, the *C-Doppelstander* served not only as a control signal but as an explicitly *non-ideological* ensign.

So, as the C flag had already served a signalling role in 1943 and carried neutral connotations, it made perfect sense in 1945 simply to cut the fly into a swallowtail and repurpose it as a temporary ensign for German shipping.

Origins and symbolic rationale — why “D” and “E”?

A similar progression unfolded in the Pacific. Just as the **C-Doppelstander** began as an improvised measure in May 1945, the same logic resurfaced with the **D-pennant** in the **Ryukyu Islands** and the **E-pennant** in Japan.

Following the **Battle of Okinawa** (March – June 1945), Admiral **Chester Nimitz** issued *Proclamation No. 1* (5 April 1945), establishing the **U.S. Naval Government of the Ryukyus** — meaning the area was already under Allied naval control by the time the battle ended in June.

On **18 July 1945**, administrative authority was briefly transferred from the Navy to the *U.S. Army Forces in the Pacific (AFPAC)*, but naval command was restored on **21 September 1945** with the creation of the *U.S. Military Government of the Ryukyu Islands*. Japanese symbols were prohibited, and civilian ships were not permitted to fly the U.S. flag. Operational evidence suggests that the *D-pennant* was already introduced locally following Okinawa's capture to mark Allied-controlled shipping during this transition period.

This timing explains the alphabetical sequence: with “C” assigned to Germany, “D” naturally followed as the next identifier once operations shifted west toward Japan.

Japan's *Diplomatic Bluebook 1967* notes that “Okinawan merchant ships adopted the ‘D’ pennant as a naval ensign immediately after the war; however, since it was not recognised, many vessels flying it were seized.” This confirms informal D-flag use before its formalisation under *USCAR Ordinance No. 148 (1955)*, which defined the ‘*Ryukyu vessel flag*’ as a double-pointed derivative of the same pennant.

Taken together, the **C-Doppelstander** (*Germany*), **D-pennant** (*Ryukyu Islands*), and **E-pennant** (*Japan*) form what appears to be both an alphabetical and chronological sequence of Allied occupation ensigns — a deliberate continuation of a proven control-signal concept. Had further zones been designated, the next logical issue might indeed have been an “F” swallowtail.

Allied occupation ensign timeline (1945):

C → Germany — May 1945
D → Ryukyu Islands — June/July 1945
E → Japan — August 1945

Alphabetical progression mirrors chronological occupation —
a consistent Allied control-flag pattern likely evolved from wartime “C” compliance usage.

Archival research is ongoing to confirm whether the C flag's control meaning was codified in Allied signal directives (1944 – 45) and to establish the precise introduction dates of the *D* and *E* pennants.

4) Surviving examples — how few, and why

Thousands must have been issued; very few survived. The duty window was short (essentially 1945–1949), the environment was harsh (salt, wind, spray), and after 1951 there was no souvenir market—many pieces were discarded or repurposed.

We conducted systematic, multilingual searches (German, English, Danish, Russian, Dutch, French, Polish, Spanish, Italian), including deliberately misleading keyword variants—e.g., “Signalflagge C,” “German flag 1945–1949,” “Doppelstander,” “double-stander,” and “Allied-occupation German flag.” Result: no public sales and almost no museum catalogue entries. When examples do surface, they are often mis-catalogued as ordinary signal flag C rather than the swallow-tailed C-Doppelstander ensign.

Requests to major maritime and historical museums yielded almost no surviving examples. However, the **Haus der Geschichte** in Bonn confirmed one preserved specimen (measured 114 × 97 cm), closely matching the Gr. 2 signal flag size class. Its existence provides a critical institutional reference alongside the privately held and published examples described below.

A rare published photo appears on page 5 of **Thilo Biegler**’s study *Die Deutschen Kriegsmarineflaggen 1935–1945* (FIAV PDF), where the C-Doppelstander is shown as “Gr. 1 C” (fig. 2). His photo confirms the signal-flag origin and depicts a tail angle visibly wider than the schematic 90°, though no exact measurements are given. Source: [Thilo Biegler – Deutschen Kriegsmarine \(PDF\)](#).

Two period photographs from the **DDG Hansa archive** show the cargo ship *Greiffenfels* flying the C-Doppelstander at the stern. The accompanying archive note states: “Als Nationalflagge wurde bis zum 23.02.1951 der Doppelstander ‘C’ gesetzt.” The photographs confirm the flag’s working use and proportions consistent with our measured pieces.

In addition, period film evidence further supports this: in the 1949 feature *I Was a Male War Bride*, a German vessel appears in the background flying the same swallow-tailed C ensign (timestamp 17:55). This provides the sixth independent confirmation of the flag’s form in actual post-war service.

Our two matching, maker-marked examples therefore form a high-confidence reference pair for this otherwise scarcely documented ensign type. For construction details, see Section 5; for measurements and ratio analysis, see Section 6.



C-Doppelstander (German occupation ensign) displayed at the stern of a merchant vessel, period photograph cited in Sections 6 and 7.

Source: © Archiv Internationales Maritimes Museum Hamburg — reproduced with permission. Original

negatives © Georg Heinemann / DDG Hansa archive, now held by IMM.



Additional *Greiffenfels* stern view showing the C-Doppelstander (German occupation ensign) in service.
Source: © Archiv Internationales Maritimes Museum Hamburg — reproduced with permission. Original negatives © Georg Heinemann / DDG Hansa archive, now held by IMM.



Film still from *I Was a Male War Bride* (1949) showing a German vessel in the background flying the C-Doppelstander (German occupation ensign).
Timestamp ≈ 17 min 55 s.

Source: 20th Century Fox — reproduced for research and documentation purposes only.

5) Observations on our two flags

Both pieces are sewn in dense wool bunting with a coarse, period weave. The headers are heavy, stiff canvas for service use and fitted with solid Inglefield-style clips; seams are neat machine lockstitch using uniform black/dark thread throughout (no thread-colour change at the joins), consistent with late-1940s German workshop practice. The fly edges show physical fraying consistent with prolonged exposure to wind and salt air, not artificial wear. On each hoist the maker's stamp "Düsseldorfer Fahnenfabrik" is clear, alongside "C" and "Gr. 2" markings.

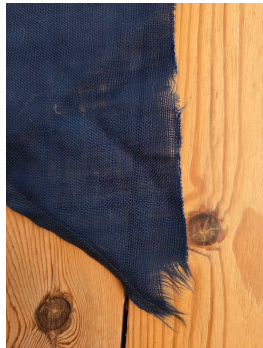
Taken together, the "C" code and Gr. 2 size mark support the working assumption that these started life as standard signal flag C blanks and were converted into C-Doppelstander by cutting a swallowtail — exactly as described in the Allied radio order ("...with a triangle cut from the fly thus transforming it into a burgee") and later codified in **Annex A** to Law No. 39.

This conversion is entirely plausible for a resource-tight post-war workshop (reuse over fresh yardage). The cut itself presents as a practical workshop template (see Section 7), not a mathematically exact 90° schematic.

This correlation between the photographs and both surviving examples indicates that makers worked to a consistent, functional pattern rather than the theoretical right angle prescribed in the regulation.

Measurements and observations are naturally approximate; any inferred geometry or ratio comparisons (Sections 6 and 7) should therefore be read with due caution. The present section records direct, physical characteristics only.

A macro of the frayed swallowtail shows the black stitching thread; the fibre sample in Section 8a was taken from this point.



FG-048 — Macro of frayed swallowtail edge on dense wool bunting, showing black lockstitch. Demonstrates honest service wear on a rare surviving example.



FG-047 — Period Inglefield-style clip on the stiff canvas header, consistent with late-1940s German merchant shipping practice.



"FG-047 — "C" code and "Gr. 2" size marking on the hoist, confirming origin from a signal flag C blank adapted into a C-Doppelstander.



FG-048 — Maker's stamp: Düsseldorfer Fahnenfabrik. Archival trail confirms workshop continuity into 1952,

6) Size and Ratio Evidence

Published summaries disagree on the base ratio of the C-Doppelstander: some give 2:3 (Imperial/Weimar precedent), others 3:5 (Nazi-period practice).

Before the measurements, a quick key to the markings on our flags: **Gr.** (*Größe*) is the German signal-flag size class — the stock workshops and ships actually had on the shelves:

Gr. 1 $\approx 50 \times 60$ cm

Gr. 2 $\approx 100 \times 120$ cm

Gr. 3 $\approx 150 \times 180$ cm

Both surviving examples are stamped **Gr. 2** and, as preserved, measure about **93 × 112 cm** (FG-047) and **96 × 115 cm** (FG-048) to the upper fly corner — i.e. much closer to $\approx 5:6$ than to 2:3 or 3:5. Their present sizes are 7–9 % under nominal Gr. 2 (100 × 120 cm), fully consistent with wool-bunting shrinkage, workshop tolerances, and in-service wear.

Two independent external examples reinforce this range. The **Haus der Geschichte, Bonn** specimen measures **114 × 97 cm** (outer dimension), again within the Gr. 2 family.

Likewise, **Thilo Biegler's published "Gr. 1 C" photograph** (FIAV paper 15-11) shows a visually comparable body ratio ($\approx 5:6$) and a broad, open wedge.

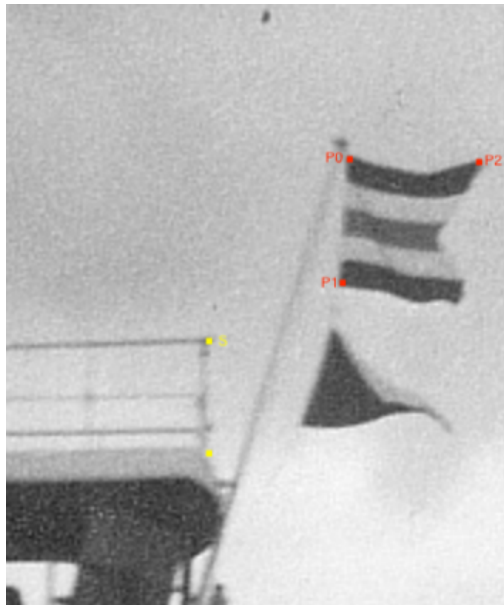
Together with our two maker-stamped pieces, the *Hansa Greiffenfels* photo, and the 1949 film still, the evidence converges on a single working template — a reused signal-flag rectangle.

In both **Allied radio order** and **Law No. 39** there were no centimetre or ratio specifications.

Bridge to 6a.

Even though the size and ratio on the six references almost speak for themselves, we ran a photogrammetric check anyway — for completeness. To cross-check the measured cloth reality against a period photograph, we add a photogrammetric estimate from the Greiffenfels stern image. The ship's railing serves as a ≈ 1.0 m scale reference, based on contemporary yard standards and prior measurements of similar vessels. Using this fixed element, we extract hoist and fly pixel lengths and correct for camera angle and wind billow. The calculation is illustrative rather than normative.

6a) Photogrammetric check (Greiffenfels, Fig. 6-1)



Reference crop from the Greiffenfels photograph, used for photogrammetric estimation.
Control lines: S = railing reference, $P_0 \rightarrow P_1$ = hoist, $P_0 \rightarrow P_2$ = fly.

Fig. 6-1: Greiffenfels stern, C-Doppelstander crop (control points as described).

Method (control points)

Let P_0 = upper hoist corner, P_1 = lower hoist corner (defines hoist $H = |P_0P_1|$).

Let P_2 = upper fly corner at the start of the swallowtail notch (where the upper edge meets the cut).

Measure along the upper edge from P_0 to P_2 to obtain the observed fly length L_{obs} .

Pixel reads (from the crop)

- Reference railing S (yellow line, ≈ 1.0 m): 83 px
- Hoist $H = |P_0P_1| \approx 91$ px
- Observed fly $L_{\text{obs}} = |P_0P_2| \approx 96$ px

Observed ratio (no correction)

$$r_{\text{obs}} = L_{\text{obs}} / H \approx 96 / 91 \approx 1.05$$

Projection and billow corrections

Two effects shorten the apparent fly length:

- (i) projection (flag plane yawed to camera) $\rightarrow$ factor $\approx \cos \theta$;
- (ii) wind billow (edge curvature; chord < arc) $\rightarrow$ factor β (≈ 0.92 – 0.96 for a moderately drawing ensign).

Correction formula

$$r_{\text{true}} = r_{\text{obs}} / (\cos \theta \times \beta)$$

Conservative bounds

$$\theta \approx 5\text{--}15^\circ \rightarrow \cos \theta \approx 0.996\text{--}0.966;$$

$$\beta \approx 0.92\text{--}0.96;$$

$$\text{Combined shortening} = \cos \theta \times \beta \approx 0.956\text{--}0.887;$$

$$\text{Therefore } r_{\text{true}} \in [1.05 / 0.956, 1.05 / 0.887] \approx 1.10\text{--}1.18.$$

Uncertainty

Pixel reads carry ± 2 px; relative to $H \approx 91$ px this is $\pm 2.2\%$.

Propagated through r_{obs} and corrections, a conservative envelope is 1.08–1.20.

Values depend on control-point placement and the 1 m rail assumption; results are indicative only.

Limitations and intent.

This analysis is illustrative and serves as a visual cross-check rather than a precision survey.

The *Greiffenfels* photo alone shows that the flag's base rectangle is visibly not 2:3.

The computed range ($\approx 1.08\text{--}1.20$) simply quantifies that observation: a reused signal-flag body adapted into a broader-tailed ensign.

Interpretation.

A base-rectangle ratio in the 1.08–1.20 band corresponds closely to 5:6 (1.20) and is well removed from 2:3 (1.50) or 3:5 (1.67). Scaled via the railing ($\approx 1.0\text{ m} = 83\text{ px}$), the hoist and observed fly correspond to $\approx 1.10\text{ m}$ and $\approx 1.16\text{ m}$, respectively. The slightly long hoist and short fly are both consistent with projection and billow effects. After correction, the effective ratio aligns with the known $\sim 5:6$ signal-flag family.

Independent check.

A repeat measurement by an independent reviewer using the same crop and method returned hoist $\approx 1.15\text{--}1.20\text{ m}$ and fly $\approx 1.22\text{--}1.33\text{ m}$ after correction. The resulting ratio again falls within the 5:6 family and well away from 2:3 or 3:5.

6b) Hypothesis — signal flag “C” as the working template

The **Allied radio order** instructed German and ex-German ships to fly the International Code Flag “C” as a **burgee** — no measurements, no ratio table. **Annex A** to Allied Control Council Law No. 39 later repeated the instruction schematically but likewise gave no centimetre data or proportional reference.

In practice, this meant that the flag’s body came directly from existing *Signal C* stock in the standard size families (**Gr. 1** $\approx 50 \times 60$ cm, **Gr. 2** $\approx 100 \times 120$ cm, **Gr. 3** $\approx 150 \times 180$ cm). The directive simply activated what workshops and ships already had on hand.

Six independent reference points support this interpretation: our two maker-stamped examples (**FG-047/048**), both **Gr. 2**; **Biegler’s** documented *Gr. 1* piece; the **Bonn Museum** flag (114 × 97 cm); the **Greiffenfels** photographs; and the **1949 film still** showing the ensign in active use. All cluster tightly around the **~5:6** ratio family rather than 2:3 or 3:5. This defines the proportions of the flag’s body. The question of the tail geometry is separate: Annex A depicts a schematic wedge, but verified examples show a visibly broader cut. Section 7b examines that divergence in detail.

7) Angles — theory vs. practice

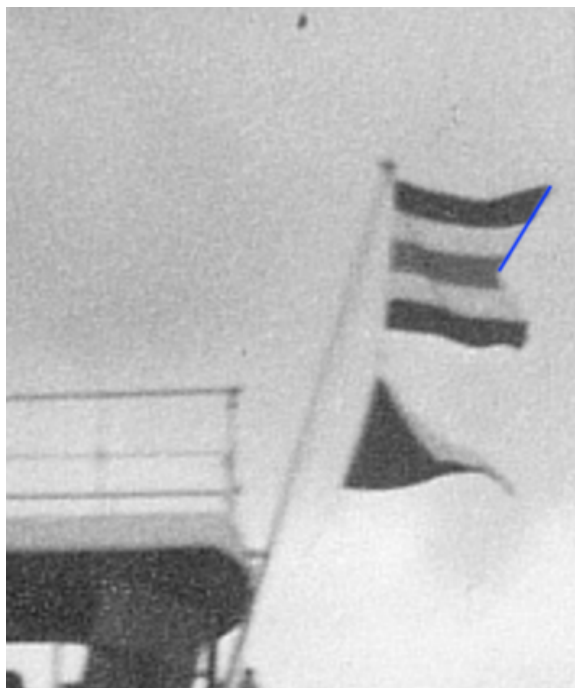
Even without calculation, the photographs and surviving flags make it clear: none of the observed swallowtails approach the 90° schematic shown in Annex A of Allied Control Council Law No. 39.

Every documented example shows a visibly broader cut—typically between **106° and 110°**.

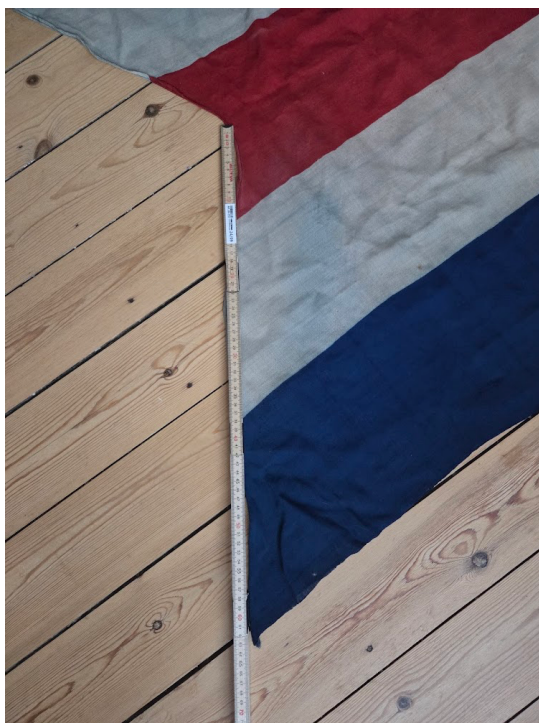
Again, just for completeness, we ran a geometric check to visualise the deviation from the theoretical 90° wedge.

We tested three independent sources:

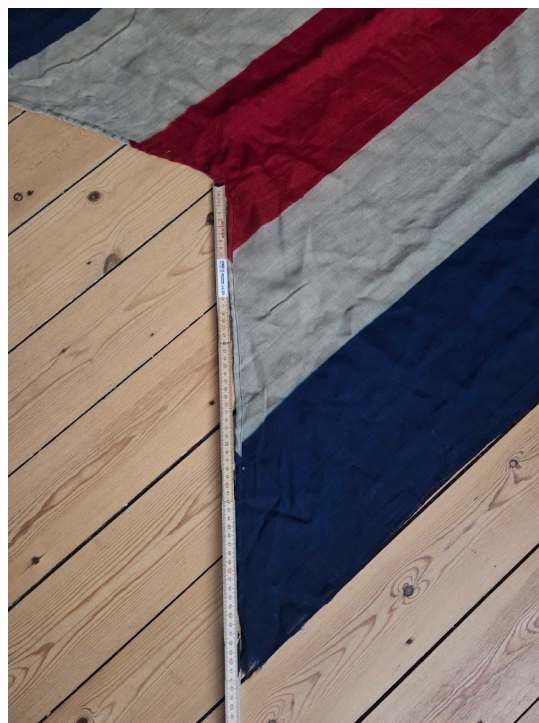
- **Hansa evidence photo**
 - Hoist (H) ≈ 1.10 m (91 px via 1 m railing)
 - Cut depth (d) ≈ 0.68 m (56 px)
 - Observed wedge angle $\omega_{\text{obs}} \approx 117^\circ \rightarrow$ corrected (projection + billow) $\omega \approx 106\text{--}110^\circ$
- **FG-047**
 - Hoist (H): 93 cm
 - Cut depth (d): 64 cm
 - Wedge angle $\omega \approx 106.3^\circ$
- **FG-048**
 - Hoist (H): 96 cm
 - Cut depth (d): 62 cm
 - Wedge angle $\omega \approx 107.1^\circ$



Reference crop from our Hansa photograph.
The blue line marks cut depth $d \approx 0.68$ m (56 px), scaled against hoist $H \approx 1.10$ m (91 px).
Used here as the photogrammetric basis for wedge angle analysis.



FG-047 — angle measurement.
Hoist $H = 93$ cm, cut depth $d = 64$ cm.
Calculated wedge angle $\approx 106.3^\circ$.

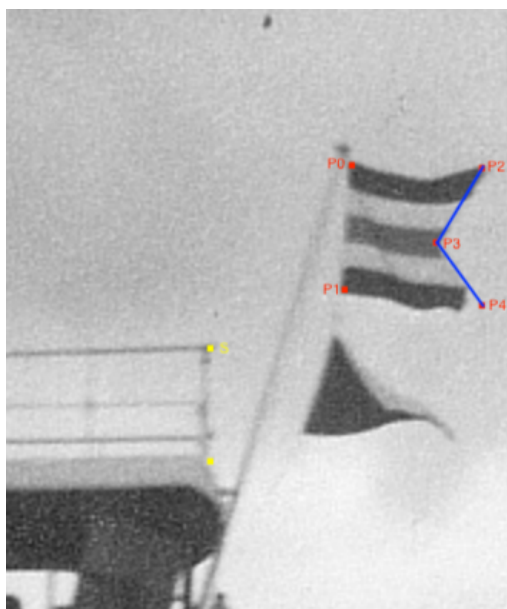


FG-048 — angle measurement.
Hoist $H = 96$ cm, cut depth $d = 62$ cm.
Calculated wedge angle $\approx 107.1^\circ$.

7a) Photogrammetric angle check (Greiffenfels, Fig. 7-1)

This subsection quantifies what the eye already suggests.

Using a single Greiffenfels stern photograph, the goal was not precision engineering but a sanity check — confirming whether the measured wedge aligns with the same ~106–110° pattern observed on both surviving cloth flags.



Greiffenfels stern crop with control points (P_0 – P_4) and scale reference (S).
Yellow = railing scale (≈ 1.0 m), $P_0 \rightarrow P_1$ = hoist, blue = cut-depth legs.

Fig. 7-1: Greiffenfels stern, C-Doppelstander crop (angle control points as described).

Method (control points)

We report the exterior (“wedge”) angle ω at the swallowtail apex.

Let P_0 = upper hoist corner, P_1 = lower hoist corner (defines Hoist $H = |P_0P_1|$).

Let P_2 = upper notch corner, P_4 = lower notch corner, P_3 = notch apex.

Let $d = \text{mean}(|P_2P_3|, |P_4P_3|)$. Compute the wedge directly as:

$$\omega = 180^\circ - 2 \cdot \text{atan}(d / H).$$

Pixel reads (from the crop)

- Reference railing (≈ 1.0 m): 83 px (S, yellow)
- Hoist $H \approx 91$ px $\rightarrow \approx 1.10$ m ($P_0 \rightarrow P_1$)
- Cut-depth legs: 57 px, 55 px $\rightarrow$ mean $d \approx 56$ px ($P_2 \rightarrow P_3$ and $P_3 \rightarrow P_4$, blue)

Observed wedge (no correction)

$$\omega_{\text{obs}} = 180^\circ - 2 \cdot \text{atan}(56 / 91) \approx 117^\circ$$

Projection & billow corrections

Perspective and sail-shape billow make the notch appear deeper; this reduces the wedge angle on the photo.

A conservative correction of -7 – 11° on ω is appropriate.

Therefore

$$\omega_{\text{true}} \approx 106\text{--}110^\circ$$

Uncertainty

± 2 px on d and H ($\sim \pm 2\%$) propagates directly; a conservative envelope is ~ 106 – 113° .

Limitations and intent.

The analysis here is likewise illustrative rather than definitive.

The *Greiffenfels* photograph alone makes it visually clear that the swallowtail is far wider than 90°, regardless of exact pixel math. The numerical estimate simply quantifies that visual reality and should be read as an indicative range, not a blueprint measurement.

Interpretation.

The exterior (wedge) angle on *Greiffenfels* resolves to ~106–110° (conservatively ~106–113°), matching the two Düsseldorf samples (~106–107°).

This consistency suggests that workshops followed a pre-existing broad-cut template — a practical continuation of earlier German naval standards rather than a literal 90° regulation. Section 7b explores this continuity further.

Peer review notes

Method: law of cosines on triangle P_2 – P_3 – P_4 , legs 55/57 px, base assumed ≈91 px (≈hoist).

Result: interior angle γ ≈109° raw; adjusted for wind/perspective ≈102–109° (mid ≈105°).

Conclusion: The cut angle at P_3 is approximately 102–109°, closest to 105° under observed conditions.

7b) Hypothesis — why the swallowtail stayed wide

Observation.

Every verified example — our two DFF flags, the *Greiffenfels* photo, Biegler's image, and the Bonn reference shows a broad swallowtail around **106–110°**.

Hypothesis.

The **9 May 1945** order simply said “fly International Code Flag C as a burgee.”

No drawings, no ratios — just a plain signal. Crews and sailmakers did what they always did: cut it by eye, using familiar swallow-tailed patterns already on board. The most likely templates (hypothesis) were the wartime **Zeppeliner pennant** (see [FOTW reference](#)) or the **A/B** patterns referenced in section 3a — later re-designated as *Bravo* and possibly *Alpha* signal flags all sharing the same broad ~105–110° wedge typical of German signal practice.

Workshop continuity.

When civilian suppliers such as *Düsseldorfer Fahnenfabrik* later produced replacement stock, they simply repeated the form already in use — the same broad tail, not the schematic right angle.

Interpretation.

The 90° drawing in Law 39 was a bureaucratic simplification of a design already fixed in practice.

By the time it appeared in print, the pattern at sea was long established — practical, visible, and cut from tradition rather than from paper.

Note:

No construction sheets have yet surfaced; this remains a working interpretation based on cloth, photographs, and workshop logic. Section 9 will return to what this implies for reconstruction and authenticity.

8) Authentication (construction × maker × service window)

But how do we know our two flags are authentic?

Here are the verifiable facts — physical, archival, and chronological — that together establish their originality and period manufacture.

Construction. Consistent with late-1940s German and Allied marine standards — dense wool bunting, reinforced canvas headers, machine lockstitch seams (uniform dark thread), and metal Inglefield-type clips. The swallowtail ensign form rules out confusion with a square signal flag C; these were purpose-made C-Doppelstander ensigns, not converted signals.

Maker timeline. Both flags carry clear “**Düsseldorfer Fahnenfabrik**” hoist stamps.

Archival evidence documents the company continuously from **1933 to 1951** under that exact name. In **1952** it reappears as *Fahnenfabrik Wunderwald* (*vormals Düsseldorfer Fahnenfabrik*), and from **1953 – 1957** as *Fahnenfabrik Wunderwald & Hammann* — before the entire business folded. This verified timeline fixes the latest possible production year for DFF-stamped flags and ties our two examples to a real, active post-war workshop.

Service window. The C-Doppelstander was in operational use from **9 May 1945** and formally in force **17 Jan 1947 – 23 Feb 1951** under Allied Control Council Law No. 39.

Synthesis. Overlaying *construction*, *maker chronology*, and *service window* leaves a single plausible interval of manufacture: a **Düsseldorfer Fahnenfabrik**-stamped **C-Doppelstander** could only have been produced during the immediate post-war occupation period — from the May 1945 order through the 1951 repeal. No later production pathway exists, supporting these as **authentic, period-made working flags**.

Independent provenance

Separate correspondence with the former British owner’s family confirms that both flags were obtained from post-war *Army Surplus* stock in East London and used to decorate a public house for the **1953 Coronation**. The description — “British supply, made in Germany for the Germans; probably never used” — aligns precisely with the known Allied surplus trade of the late 1940s and early 1950s.

This account provides independent confirmation that Düsseldorfer Fahnenfabrik occupation flags circulated in civilian hands by the early 1950s, consistent with the workshop’s closure window.

*“My family have run pubs in East London for over 100 years...
These flags came from one of those stores — probably never used.
It was bought before 1953 to decorate a pub for the Queen’s Coronation.
Though not a British flag, anything red, white and blue will do.”*

— Peter, former pub owner (London), personal correspondence, 2025

The precise route by which these flags reached the United Kingdom remains unknown, as does the specific vessel or depot from which they originated.

Materials control (lead-in to section 8a).

To rule out a later conversion of a square signal flag C into a C-Doppelstander, a **fibre burn test** was performed on the exposed black stitching thread of FG-048’s swallowtail; results follow in **section 8a**.

8a) Fibre burn test (FG-048 swallowtail — black thread)

Purpose. To demonstrate that the swallowtail cut and its stitching are period-executed (late-1940s/early-1950s) rather than a later conversion of a 1930s–40s stock signal flag “C” (or a modern repair). Synthetic threads (polyester/nylon) melt and bead; period natural fibres ash.

Sample. Single strand taken from the black stitching thread visible in the macro close-up at the frayed swallowtail edge of FG-048.

Method. Handheld flame; behaviour noted before ignition, during burning, and at residue.

Follow the sequence: 1 → 2 → 3 → 4 → 5



1. Overview of the swallowtail area on FG-048. The sample thread was taken from the exposed stitching at the notch.



2. Macro close-up showing the black stitching thread at the frayed edge (sampling point).



3. Thread sample placed on plate *before* ignition.



4. After burn: macro of the friable ash residue (no hard, glassy bead as with thermoplastics).



5. Ash crushed to powder on fingertip—behaviour of natural fibre (e.g., wool), not synthetic.

Observations

- No melting or beading; no shrink-back.
- Brief yellow/orange flame, then char.
- Residue: soft, friable grey-black ash (crushes to powder; no hard glassy bead).
- Odour: organic “burnt cloth/hair”; no sweet/chemical note.

Interpretation

- Behaviour is diagnostic of natural fibres and incompatible with modern thermoplastics.
- Odour/ash is consistent with proteinaceous fibre (wool), matching the flag’s wool bunting and late-1940s workshop practice for sewing threads.
- **By eye, the black stitching at the notch matches the reinforcement-thread on the header in twist, diameter and colour.** While not a laboratory proof, this visual match supports that the swallowtail hemming was executed in the same workshop session — consistent with *factory* work at DFF rather than a later, off-site conversion.
- Together with the uniform dark thread seen across seams in section 5, this supports an in-period tail cut and stitching, not a later add-on or post-war “upgrade” from a square signal flag.

Limitations. Field test only; microscopy/FTIR would discriminate wool vs. cotton definitively. For our purpose—screening for modern synthetics—the result is decisive.

Conclusion. The black swallowtail thread on FG-048 behaves as a period natural-fibre thread (consistent with wool) and does not show synthetic signatures, arguing against a later conversion of an ordinary signal flag “C” into a C-Doppelstander.

Supplementary note (provenance cross-check).

A follow-up exchange with the former owner in 2025 further supports the integrity of the flags’ original shape. When asked whether the flags had ever been modified, he replied that he had “only ever known them as swallow-tailed” and had “no reason to think they were ever altered.”

This recollection aligns with the physical and material evidence that the triangular cut was executed in-period, not added later for decoration.

9) Summary and conclusion

(1) Authenticity.

Both flags are confirmed period-made. Their *Düsseldorfer Fahnenfabrik* stamps, materials, thread types, and construction details align perfectly with verified late-1940s practice.

Each parameter — maker, markings, fibre, geometry — matches the four external references (Bonn, Biegler, *Greiffenfels*, and film stills). Together they form a coherent six-point dataset confirming genuine, working C-Doppelstander ensigns.

(2) Ratio and size.

All six reference points cluster near **5:6**; no evidence supports 2:3 or 3:5.

Both the **9 May 1945 war diary** and **Law No. 39** confirm the use of signal C as the base form, with Annex A adding only a schematic sketch. The right-angle shown there appears purely illustrative — no surviving flag or photo matches it.

(3) Origin and manufacture.

Workshops and ship crews followed the 9 May 1945 instruction literally: fly a *signal C burgee*.

They simply cut existing signal-flag stock to the familiar swallow-tail used on wartime pennants (Zeppeliner/Alpha/Bravo). Later production by DFF followed the same logic, yielding the same broad $\approx 106\text{--}110^\circ$ wedge.

(4) Why “C” — and the link to D / E.

Archival leads suggest that **Signal C** was already used during 1943 as an *Allied control identifier* hoisted below national ensigns, and was later reused by Germany after the surrender as a standalone modified ensign. This also served a symbolic purpose — adopting a *neutral signal flag* rather than any national design with Nazi associations.

Law No. 39 justified the swallowtail form “to avoid confusion” with the square signal, fully consistent with this reuse. The C-based configuration therefore became the working template for the later **D** (Ryukyu) and **E** (Japan) occupation flags.

Despite popular claims, there is **no evidence** that the “C” stood for “Control,” or that its colours were intended to represent Allied forces or national ensigns. Both notions remain unsubstantiated by any primary source or regulation.

In summary:

The C-Doppelstander was a practical, workshop-made ensign — ratio $\approx 5 : 6$, wedge $\approx 106\text{--}110^\circ$, signal-flag materials — created directly from existing stock under the 9 May 1945 order and maintained until 1951. Based on archival findings, it is clear that when U.S. and British forces took control over enemy territories, they routinely employed reconstructed *International Code of Signals* flags as neutral merchant ensigns in place of banned national flags. This practice explains both the reused signal-flag proportions ($\approx 5 : 6$) and the absence of any verified $2 : 3$ or $3 : 5$ variants. No confirmed examples show a strict 90° cut, reinforcing that the surviving flags represent the real, working form.

This study also sets the stage for a forthcoming paper exploring the **broader Allied use of signal flags** — tracing how the **A, B, C, D, E** (and possibly **F**) swallow-tail series evolved from wartime communication standards into post-surrender control and identification flags across multiple occupation zones. The next phase will ask whether this sequential system was coincidence, field improvisation, or a deliberate Allied framework.

Sources / references

Section 3 – The C-Doppelstander: legal and practical use

- Allied Control Council Law No. 39 (issued 12 Nov 1946; in force 17 Jan 1947; withdrawn 23 Feb 1951). *Amtsblatt des Kontrollrats in Deutschland*, 1946, Nr. 12, S. 226. Annex A shows the swallowtail conversion of signal flag “C” and notes a 90° cut (no dimensions/ratio).
PDF: [Amtsblatt ... 1946 Nr. 12 S. 226 \(Annex A\)](#)
- Royal Navy War Diary, May 1945, Part I (1–15 May). Entry of 9 May 1945 ordering German vessels “to fly at the peak International Code Flag C with a triangle cut from the fly thus transforming it into a burgee.”
[Royal Navy Historical Branch \(PDF, p.215\)](#)
- Wikipedia entry: *Erkennungsflagge für deutsche Handelsschiffe* (overview of usage and regulation context).
de.wikipedia.org
- Flags of the World (FOTW): German merchant ensign 1945–51, incl. notes on C-Doppelstander.
crwflags.com

Section 3a – Why “C”?

- *Combined Chiefs of Staff Signal Summary*, Vol. IV (August 1943) — preparatory **test broadcast** drafted during the Italian armistice planning, instructing surrendered Italian ships to hoist the International Code Flag C beneath their national ensign under Allied supervision.
[National Archives and Records Administration \(NARA\), Record Group 38, image 59.](#)
- British Library, Additional Manuscripts 71786 ff. (1946–47) — British Naval Control Service correspondence referencing the inspection of surrendered German ships and the issue of replacement ensigns and authorised flag sets (Kiel, Flensburg, Bremerhaven).

Section 4 – Surviving examples

- DDG Hansa — *Greiffenfels* with the C-Doppelstander (German occupation ensign) at the stern; archive note: “Als Nationalflagge wurde bis zum 23.02.1951 der Doppelstander ‘C’ gesetzt.”
[greiffenfels223.jpg](#) (© Georg Heinemann)
- *I Was a Male War Bride* (20th Century Fox, 1949) — film still at ≈ 17 min 46 s showing a German vessel in the background flying the C-Doppelstander (German occupation ensign).
[Film still \(research extract\)](#)
- **Haus der Geschichte, Bonn** — one preserved C-Doppelstander, measured **114 × 97 cm** (outer dimensions). Confirms a Gr. 2 size class consistent with our two specimens. The museum has confirmed these details and provenance (Marineortungsschule Bremerhaven, 1991 acquisition) in direct correspondence. Communication on file; not reproduced here in full.
- A period illustration by **Thilo Biegler**
([International Federation of Vexillological Associations publication](#))

Section 6 – Size and ratio evidence

- Documented period signal-flag sizes encountered: Gr. 1 ≈ 50×60 cm, Gr. 2 ≈ 100×120 cm, Gr. 3 ≈ 150×180 cm (incl. pilot/“Delta” examples). These underpin the Gr. 2 logic relative to our measured pieces.
[Militaria Plaza, Gr. 1 ≈ 50×60 cm](#)
[Ratisbons.com, Gr. 2 ≈ 100×120 cm](#)
[IMA USA, Gr. 3 ≈ 150×180 cm](#)
- **Haus der Geschichte, Bonn** — one preserved C-Doppelstander, measured **114 × 97 cm** (outer dimensions). Confirms a Gr. 2 size class consistent with our two specimens. The museum has confirmed these details and provenance (Marineortungsschule Bremerhaven, 1991 acquisition) in direct correspondence. Communication on file; not reproduced here in full.
- A period illustration by **Thilo Biegler**
([International Federation of Vexillological Associations publication](#))

Section 7 – Angles: theory vs. practice

- *Flaggenbuch* 1893 — Imperial naval swallowtails shown with open wedges clearly exceeding 90°. [DFG Viewer](#)
- *Flaggenbuch* 1928 — Weimar naval swallowtails again depicted with wide, open fork cuts (~105–110° visual estimate). [Universitäts- und Landesbibliothek Düsseldorf](#)
- Flags of the World (FOTW): Zeppelin pennant, Nazi period, illustrating a swallowtail cut wider than 90° (visual comparison only, no official table located). [crwflags.com – Zeppelin pennant](#)
- DDG Hansa — *Greiffenfels* with the C-Doppelstander (German occupation ensign) at the stern; archive note: “Als Nationalflagge wurde bis zum 23.02.1951 der Doppelstander ‘C’ gesetzt.” [greiffenfels223.jpg \(© Georg Heinemann\)](#)

Section 8 – Authentication / maker evidence

- *Adreßbuch für die Landeshauptstadt Düsseldorf* 1934 — entry at Hermann-Göring-Straße 102 (Nazi-era name for Friedrichstraße; same building). [ULB Düsseldorf – Digitale Sammlungen](#)
- *Adreßbuch* 1942 — Friedrichstraße 102 persists. [ULB Düsseldorf – Digitale Sammlungen](#)
- *Branchenbuch* 1948 — Fahnenfabrik A. Wunderwald, Friedrichstraße 102 (post-war activity). [ULB Düsseldorf – Digitale Sammlungen](#)
- *Adreßbuch* 1952 — “Fahnenfabrik Wunderwald (vormals Düsseldorfer Fahnenfabrik)”, Friedrichstraße 102 (explicit “vormals” bridge). [ULB Düsseldorf – Digitale Sammlungen](#)
- **Note:** 1947–1951 address books are not available online; the timeline is reconstructed from interwar/wartime books, 1948 *Branchenbuch*, 1952 address book, and register data.
- *Reichs-Telefonbuch* 1938 — listing for Düsseldorfer Fahnenfabrik A. Wunderwald, Friedrichstraße 102. [digi-hub.de viewer](#)
- Handelsregister AG Düsseldorf HRA 116448 — Fahnenfabrik Wunderwald & Hammann (partners Alfred Wunderwald / Herbert Hammann), founded 23-02-1953, dissolved 1956, deleted 1957. [Online-Handelsregister](#)
- Handelsregister HRA 111248 — Werkstätten Alfred Wunderwald. [Northdata summary](#)
- Company postcard (23 Aug 1933) — Düsseldorfer Fahnenfabrik A. Wunderwald, Benrather Straße 18 (confirms pre-war operation and branding). [lot-tissimo.com auction listing](#)
- Collectors’ market — surviving Third Reich banners stamped Düsseldorfer Fahnenfabrik (various auction listings, 2000s–2020s), corroborating the workshop’s real-world output. [Militaria Plaza](#)
[About WW2 Militaria](#)

Section 8 / provenance

- Seller correspondence (2025): East London publican family; both flags bought as army surplus in London and used as pub decoration for the 1953 Coronation and later jubilees/funerals; no military claims made.

Acknowledgements

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